

9	D	At constant speed, engine force = resistive force rate at which energy is delivered = rate at which energy is dissipated $P = Fv$ $12 \times 10^3 = F \left(\frac{72 \times 10^3}{60 \times 60} \right)$ $F = 600 \text{ N}$ $E_{\text{from fuel}} = E_{\text{to car}}$ $(0.30)(40 \times 10^6)(m) = (12 \times 10^3)(1 \times 60 \times 60)$ $m = 3.6 \text{ kg}$
10	C	The minute hand takes 1 hour to go round the clock once